

From Curves to Cascades: Composed Coupling Prediction Among Autonomous Agents

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Abstract

Autonomous agents that share a substrate couple through it, and prior work established that the resulting edges can be detected and their response curves measured — pairwise, with error bars, while the system is live. This paper asks the next question: do those local measurements *compose*? Can curves measured on neighboring links, chained through nonlinear switching elements and summed where paths converge, predict far-end behavior without walking the whole path? The referee design keeps the test honest: the pieces that compose into a prediction and the direct walk that judges it are disjoint measurements, and the comparison is made against propagated error bars built from the local noise of each piece. Across the campaign — four nonlinear chain seeds, a converging junction, and a live discontinuity relay across three seeds — 95 of 97 referee points fall within propagated two-sigma bars, and the propagated bars are themselves validated: the hit rate matches the nominal rate without over-rejection. The measurement horizon is measured, not assumed: roughly two nonlinear hops at noise $\sigma=0.02$, beyond which the far end falls below the detection floor and composition stops being an economy and becomes the only remaining door. Boundary rows (a dead planted cell, a starved-scheduler pair) are reported as receipts, not discarded.

Keywords: active probing, autonomous agents, multi-agent systems, coupling discovery, composition, response curves, uncertainty propagation, cascades, system identification, live systems, complex systems

1 Introduction

When autonomous agents tend the same substrate, each one’s parameter changes travel through the substrate into the others’ objectives. The coupling is real, hidden, and — as a line of prior work has established — measurable without central observers and without disrupting the system: edges are discovered by guardrail-bounded probing between the agents themselves [8], and each edge yields a response curve with per-point error bars, measured by a self-directed characterization walk [7].

A map of measured edges answers local questions. The operational questions are not local. *If this agent pushes its parameter, what happens two hops away? Which cascade paths are live, and where will a change arrive first? How much of a far-end shift is attributable to this sender?* Answering any of them requires the multi-hop response — and measuring it directly runs into a wall that is arithmetic, not engineering: coupling attenuates multiplicatively along the path while noise does not, so the far end of a deep chain sinks toward the noise floor. This paper’s boundary experiment shows the wall in numbers: a three-hop nonlinear chain composes to roughly 0.001 of the source’s span, twenty times below the $\sigma=0.02$ floor, and the direct walk there measures faithful flat noise (§6). Far-end measurement is not slow; past the horizon it is *vacuous*.

The alternative is to measure neighbors and compose. One hop away, the signal is strong, the bars are thin, and the measurement is cheap. If the measured curves compose — if the response of a relay

can be read off a neighbor’s self-map, if converging paths sum, if a discontinuity predicted from pieces lands where the direct walk later finds it — then the reach of the map is the reach of arithmetic rather than the reach of the probe. And if the error bars compose with the curves, the prediction carries its own price tag.

This paper tests exactly that, and the contributions are the clauses of the test:

1. **Composition.** Pairwise-measured coupling curves, chained through nonlinear switching elements (“doors”) and summed at a converging junction, predict far-end responses without walking the path: 95 of 97 referee points within propagated two-sigma bars across chains, junction, and discontinuity rows (§6).
2. **Metrology.** The error bars compose too. Local per-point bars, propagated through the composed stages in quadrature with local slopes, neither over-reject (the strict test that ignores the composed side’s uncertainty loses four more points) nor dissolve into vacuity (median referee deviation $\approx 0.3\sigma$). The composed map prices its own predictions (§4).
3. **Discontinuity.** Composition survives a live threshold: the predicted far-end step matches the directly measured step (-0.159 predicted against -0.161 planted, the direct walk landing on both), and the flip is bracketed between the same two levels the pieces pin down — across three seeds (§6).
4. **Boundary.** The composition horizon is measured, not assumed: ≈ 2 nonlinear hops at $\sigma=0.02$. Beyond it the referee itself goes vacuous — the honest outcome, reported as a result (§6).
5. **Certification.** The two rows run under the final protocol — fair lease turn-taking with a visible turn ledger — complete all four walkers’ maps in a single cell each, with rotation verified event by event. The scheduler defects that earlier rows survived are banked as disclosed before-receipts, not hidden (§6).

The framing is deliberately plain: one method, measured boundaries. Cascade identification through known blocks is a classical subject; what is new here is the specific assembly — distributed, autonomous, behavioral measurement whose pieces compose with priced uncertainty on a running system. §2 says the classical part first. All runs carry external identifiers; every figure regenerates from the retained evidence by a shipped script.

2 Related Work

Block-oriented system identification. Composing a nonlinear cascade from measured blocks is textbook material in the Wiener–Hammerstein lineage [9, 2]: a system is modeled as static nonlinearities interleaved with linear dynamics, and each block is identified separately, then composed. We state this first, before a reviewer does: nothing in this paper’s composition arithmetic is new. What differs is the measurement condition. Classical block identification assumes one experimenter who owns the plant, can isolate signals, and pauses the world between experiments. Here the “blocks” are couplings between autonomous agents that neither own nor see each other; every reading is attributed by protocol (turn-taking, parking, block design) rather than by experimental isolation; and the composed prediction is checked against a second, disjoint measurement taken by the same protocol on the same running system. System identification [3] remains the mature neighbor for the single-owner quiet case. Passive methods [6] cannot see through self-generated exploration noise at realistic budgets, as the detection study established [8].

Sharing a substrate fairly. The measurement protocol serializes probing through leases; when several walkers contend, the scheduler question is fair sharing. The turn ledger used here descends

from fair queueing — deficit round-robin [5] and the max-min lineage behind it [1] — instantiated at equal weights on lease acquires instead of packets. §6 reports both sides of this: rows taken under a defective scheduler, and certification rows taken under the repaired one.

Empirical coupling discovery. The space this line of work occupies — multiple autonomous experimenters discovering the coupling structure of the live substrate they act on, with certified quantitative output — has, to our knowledge, no other occupant. The relation to causal inference [4] is parallel rather than subordinate: Pearl’s experimenter is centralized and hypothesizes a model; here the experimenters are the coupled nodes themselves, and the model is measured edge by edge. No novelty claim is made beyond the assembly.

3 The Instrument, Briefly

The instrument is unchanged from prior work; the question is new. This section fixes vocabulary. The substrate is a set of nodes, each mapping parameters to objectives through a base function (linear, polynomial, threshold, saturation); directed edges feed one node’s (noisy) channel value into another’s, scaled by a planted strength. Autonomous agents — *breeders*, one per node — walk their own parameter axis under a push/hold/pause block protocol coordinated by fencing-token leases: the sender probes a level, receivers park and hold still, the pause block checks reversibility. Readings are block medians with error bars from local noise (median absolute deviation); no response model is fitted anywhere. A self-directed walk chooses its next level inside the gap where the most structure could still hide, and retires under a priced stop when every remaining gap could not hide more than the measurements are uncertain by [7].

Two properties matter for composition. First, curves are *level-addressed*: a point’s address is the total input level at the measured node, not the knob position that produced it. Second, every walk banks curves from *all* receivers, so the map of each sender is complete at retirement — including the sender’s own self-curve. Both properties earn their keep below; the one referee miss in the discontinuity family is a single read taken while a bystander’s own dial was off its park position — a dial-addressed lookup mis-attributing a correct reading, which level addressing tolerates.

Multi-walker cells add the scheduler. The final protocol keeps a turn ledger (per-group acquire counts) and grants the lease to the walk-pending contender with the fewest turns; certification rows run under it (§6.4). Its lineage is fair queueing [1, 5].

4 The Composition Claim and Its Referee

4.1 The claim

Let each measured curve be a set of triples (x, y, σ) : input level, response, bar. The composition claim is that for a path $n_1 \rightarrow n_2 \rightarrow n_3$ through nonlinear elements, the response at n_3 to n_1 ’s level L equals the tail map evaluated at the level induced by the first hop,

$$\hat{y}(L) = S_3(\phi(P(L))), \quad (1)$$

where P is the measured $n_1 \rightarrow n_2$ piece, S_3 the measured self-map of n_3 , and ϕ the level algebra that converts the piece’s output into the tail map’s input address — in the discontinuity scenario, $\phi(u) = 50 + 45u$: a channel addition of $0.45u$ at a node parked at 0.5 lands at the self-map’s level $50 + 45u$. At a converging junction the two path outputs add in the channel before the tail map is evaluated. The algebra is elementary; its discipline is *level is level* — addresses are total input levels, never knob positions.

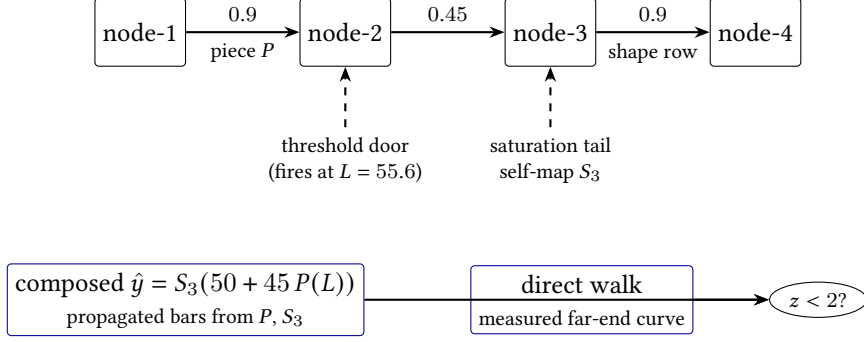


Figure 1: The referee pattern (discontinuity scenario). Pieces measured by the walkers compose through the level algebra into a far-end prediction; the same walk’s direct curve judges it. Pieces and referee are disjoint measurements; the comparison is priced by propagated bars (3).

4.2 The referee

Equation (1) is checkable because the map contains its own control: the sender’s walk banks a *direct* curve to the far end alongside the pieces. The referee test compares the composed prediction against this direct walk, point by point:

$$z(L) = \frac{|y_{\text{direct}}(L) - \hat{y}(L)|}{\sqrt{\sigma_{\text{direct}}^2(L) + \sigma_{\text{pred}}^2(L)}}, \quad z < 2 \text{ certifies.} \quad (2)$$

The design has three teeth. *Disjointness*: the pieces and the referee are different measurements; a defect in either fails the test (the starved-scheduler rows below show exactly this). *Priced uncertainty*: σ_{pred} is propagated from the pieces — each stage contributes its local bar and its local slope acting on the input’s bar, combined in quadrature,

$$\sigma_{\text{stage}}^2 = \sigma_{\text{map}}^2(x^*) + (s_{\text{map}}(x^*) \sigma_{\text{in}})^2, \quad (3)$$

with bars taken at the bracketing measured levels (the flanks’ maximum) — conservative by construction. *Planted truth*: the bench’s closed-form responses score every point against ground truth the agents never see, so piece fidelity and shape are separately auditable. The referee rule (2) is the single rule applied to every row of this paper; where a stricter variant (direct bar only, treating the prediction as exact) changes a count, the change is reported (§6.6).

Figure 1 fixes the pattern for the discontinuity scenario: the relay piece and the tail self-map compose into the far-end prediction; the direct walk judges it.

5 Experimental Design

The bench as oracle. All rows run on the parameterizable substrate of prior work [7]: planted topologies, closed-form responses, Gaussian noise $\sigma=0.02$ per reading, truth invisible to the agents. Scenarios are chosen to stress one composition structure each: *door-chain* (a nonlinear door between two saturating stages, four seeds), *junction-gate* (two converging relay paths into one tail node), *cliff* (a threshold relay mid-chain whose firing creates a live discontinuity, three seeds), *poly* (smooth convex nonlinearity, shape validation), and *depth4* (three cascaded nonlinear stages — the boundary probe). Table 1 lists run identifiers.

Pre-flight gate. Before any dispatch, the planted far-end span must exceed several noise bars, and every door’s switching condition must be reachable through the upstream output range. Both checks are one-liners over the planted forms; the campaign’s one dead cell (§6.5) is the receipt that motivates them — authored before the gate existed.

Table 1: The composition campaign. Referee = direct walk vs composed prediction, rule (2); “pts” counts referee points. Run identifiers are external (GitHub Actions run ids); † = pre-repair scheduler (disclosed, §5).

Row	Run(s)	Referee	Notes
Linear 3-hop precedent	chain4 bridge cell	—	pred 0.0956 = measured 0.0956 vs planted 0.105
Nonlinear chain, 4 seeds	door-chain s42–45	28/28 pts	two sequential doors
Junction (converging)	33522428904	11/12 pts	max $z=2.10$ at $L=0$; step exact
Live discontinuity	33623243676	33/34 pts	miss = one off-park read; piece 34/34
certification, s47	33968683471	11/11 pts	piece 11/11, shape 11/11; 4/4 walks
certification, s48	33974815211	12/12 pts	piece 12/12, shape 12/12; 4/4 walks
Poly shape validation	33607018911	12/12, 12/12	worst z 0.83/0.95; spans $30\sigma/18\sigma$
Dead-plant salvage	33597128644	—	piece 18/18; 36 pts no phantom firing
Depth boundary	33558088446	vacuous	3 hops compose to 0.001 = $20\times$ below floor
<i>Before-receipts (scheduler defect, excluded from certification)</i>			
Junction, starved	33634967561 [†]	20/22	tail self-map 3 pts; misses localized
Junction, starved	33644675248 [†]	14/14 (luck)	same defect, clean by chance; excluded
Junction, fair-but-thin	33673988909	25/31	turn ledger live, self-map still thin (see text)

Scheduling, disclosed. Mid-campaign, multi-walker rows exhibited a lease-starvation defect: one walker monopolized probe windows and a tail node’s self-map retired at three points. The defect and its fix (the turn ledger of §3) are part of the record: affected rows are reported with their verdicts *and* their map depths, the fair-scheduler repair is certified by its own rows, and the two certification cells of §6.4 run the final protocol end-to-end. Runs from before the repair are marked in Table 1.

Doctrine. As in prior work [7]: no minimum performance bar; expected degradation at the map’s edge is a result, not a failure; the only defect signal is the strong-signal/low-noise corner contradicting itself. Verdicts are recomputed from retained curves, never read from run badges; verdict tooling is reviewed like engine code (one early verdict pass had a unit bug that mis-scored a row until repair — the receipt is in the run record).

6 Results

6.1 Chains: the anchor rows

The door-chain rows are the anchor. Four seeds walk a chain whose intermediate node is a genuine nonlinear door between saturating stages; the referee compares the composed two-hop prediction against the direct walk at every measured level. All 28 referee points across the four seeds certify within propagated bars; the linear precedent row bookends the family from the other end — three linear hops whose composed prediction (0.0956) matches the directly measured far-end response (0.0956) against a planted 0.105. Composition holds through smooth nonlinearity, through switching nonlinearity, and in the linear limit.

6.2 The junction: composition under superposition

Where two paths converge (Figure 2), the composed input to the tail node is the *sum* of the two path outputs — a saturation relay on one path, a threshold relay on the other — and the tail’s own self-map converts the summed channel level into the far-end response. The referee certifies 11 of 12 points; the single miss sits at $L=0$ with $z=2.10$, on the flat shelf where both relays are off. Two features deserve the reader’s eye in the figure. The junction map (lower left) is measured by the tail node walking *its* own dial — the composition reuses it through the level algebra, never re-measuring the far end. And

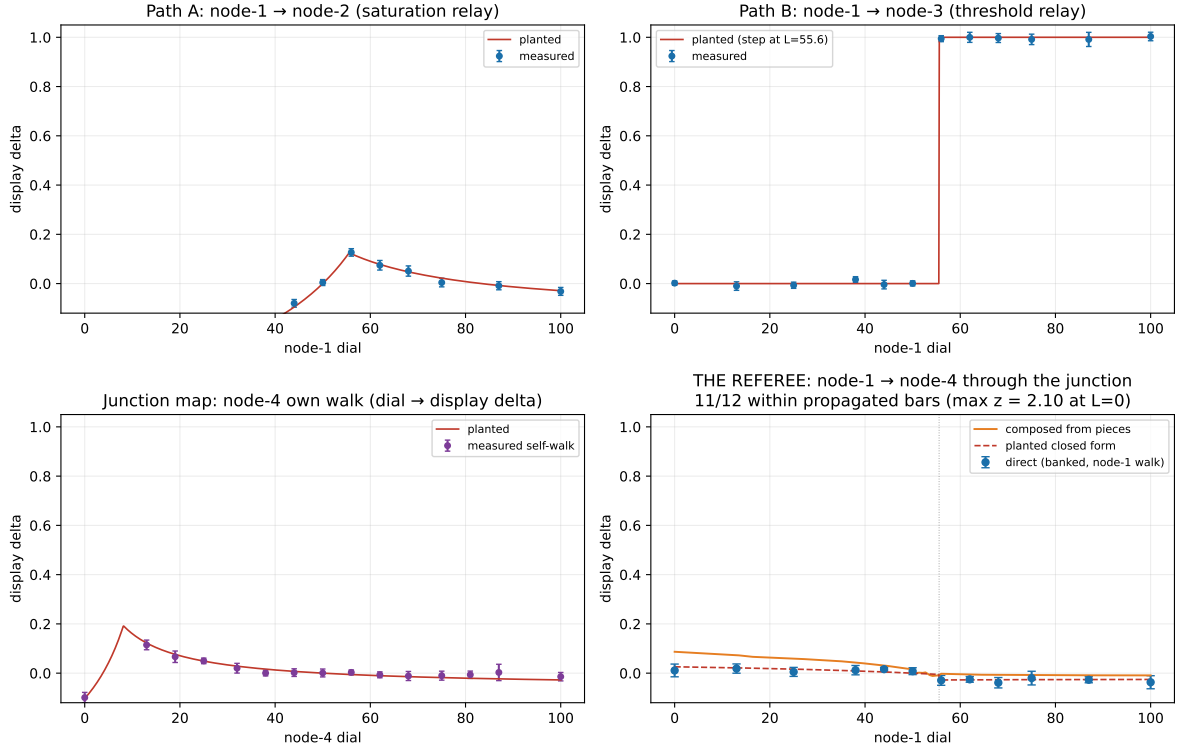


Figure 2: Junction row (run 33522428904, seed 46). Top: the two relay pieces. Bottom left: the tail node’s self-walk, reused through the level algebra. Bottom right: the referee — direct curve (blue), composed prediction (orange), planted closed form (dashed red), shared y -axis: the noise floor is one floor.

the predicted step lands exactly where the direct walk finds it: the threshold path fires at the planted $L=55.6$, and both prediction and direct curve step there together.

6.3 The cliff: composition through a live discontinuity

The cliff scenario places a threshold relay mid-chain: below the firing level the far end sees nothing; above it, the tail saturation takes the full relayed step. Composition here is not interpolation — the prediction must reproduce a jump *it has never seen at the far end*, using only the relay piece (measured one hop in) and the tail self-map (measured by the tail node itself).

Figure 3 shows a certification cell (seed 47). The relay piece pins the firing bracket between levels 53 and 56; the composed prediction steps from the flat shelf to -0.151 across the same bracket; the direct walk lands on it at every level — 11/11 within propagated bars, far-end step measured at -0.158 against a planted -0.161 . Seed 48 (Figure 4) repeats the cell under the final protocol: 12/12, step -0.163 , flip bracketed one level tighter (54–56). The earlier discontinuity row (33/34) carries the family’s instructive miss: at $L=50$ the direct read is -0.4995 — exactly the tail node’s self-curve bottom — because one sample was taken while the tail’s own dial sat near zero instead of parked. The reading was *correct* at its true input level; the curve’s dial-addressed label was wrong. The level-addressing discipline of §3 is the fix, and the miss is the receipt that motivated it.

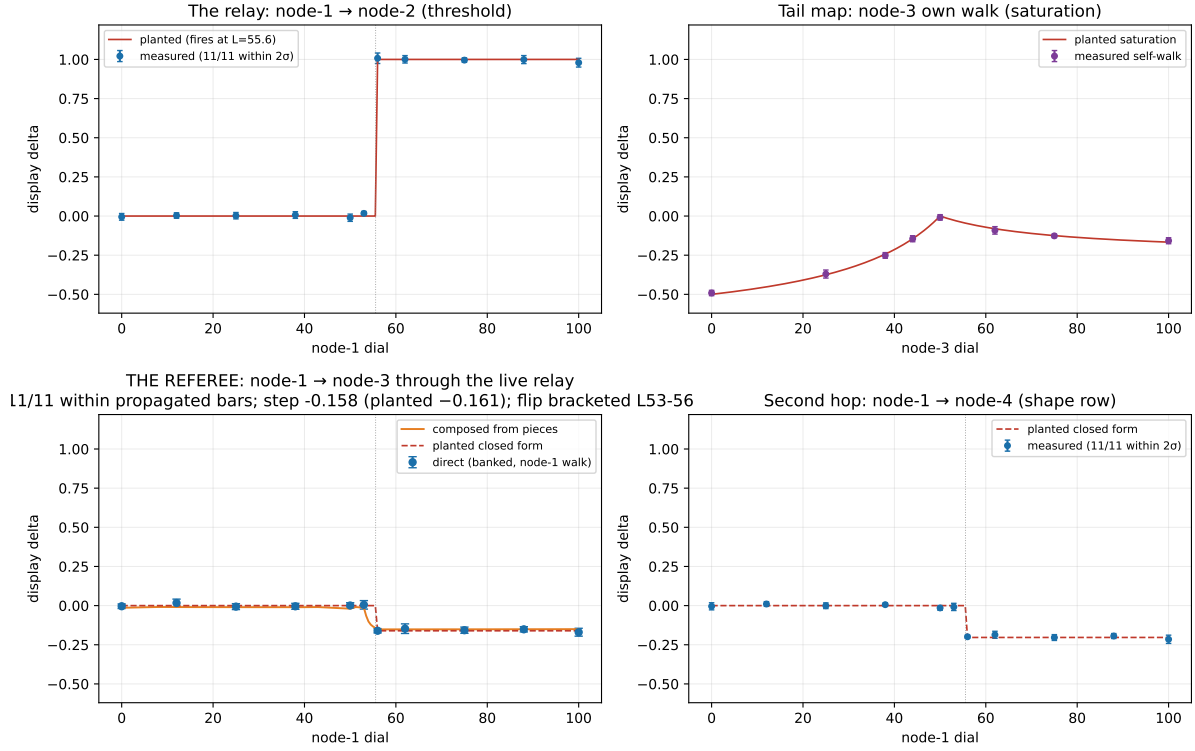


Figure 3: Certification cell, seed 47 (run 33968683471). The relay piece and tail self-map compose into the far-end prediction (lower left); the direct walk judges it. Shape row (lower right): second-hop response against planted closed form.

6.4 Certification cells: the protocol receipt

The two cliff certification cells are the campaign’s protocol rows, run end-to-end on the final build: fair turn-taking with the visible turn ledger, no timing overrides, terminal close. Both cells complete all four walkers’ maps — the first four-walker cells in the campaign’s history to do so — with every curve retired by the priced stop, and rotation verified event by event from the coordination trail: 13 lease acquires against 10 quantum yields and 22 denials (seed 47), strict walker alternation, handovers completing in 3–7 seconds, the waiting walker winning every handover, and every denial correct (an ex-holder arriving early is refused at the door). The priced stop’s depth allocation is visible in the same record: the source’s walk retires at 11 levels, the tail’s at 8, and the node whose dial moves nothing (planted: no outgoing edges) retires at 6 points of correct flat — its six points sit within 1.1σ of planted zero. A complete four-walker cell costs ≈ 2 hours wall-clock.

The scheduler receipt runs through the whole campaign and is worth stating as one line of method: verdict quality under a defective scheduler is a lottery — the starved seed 48 row passed 14/14 *by luck* while the identically starved seed 47 failed 20/22 with both misses localized to the coarse map region — and the repair is certified by its own rows, not asserted. The fair-but-thin row (25/31) then isolated a second defect (the retirement verdict ignoring the walker’s own self-curve), whose repair preceded the two certification cells above.

6.5 Shapes, salvage, and the boundary

Three rows complete the campaign’s edges. The *poly* rows validate shape composition on smooth convex nonlinearity: polynomial piece and saturation-of-polynomial, 12/12 and 12/12 against planted

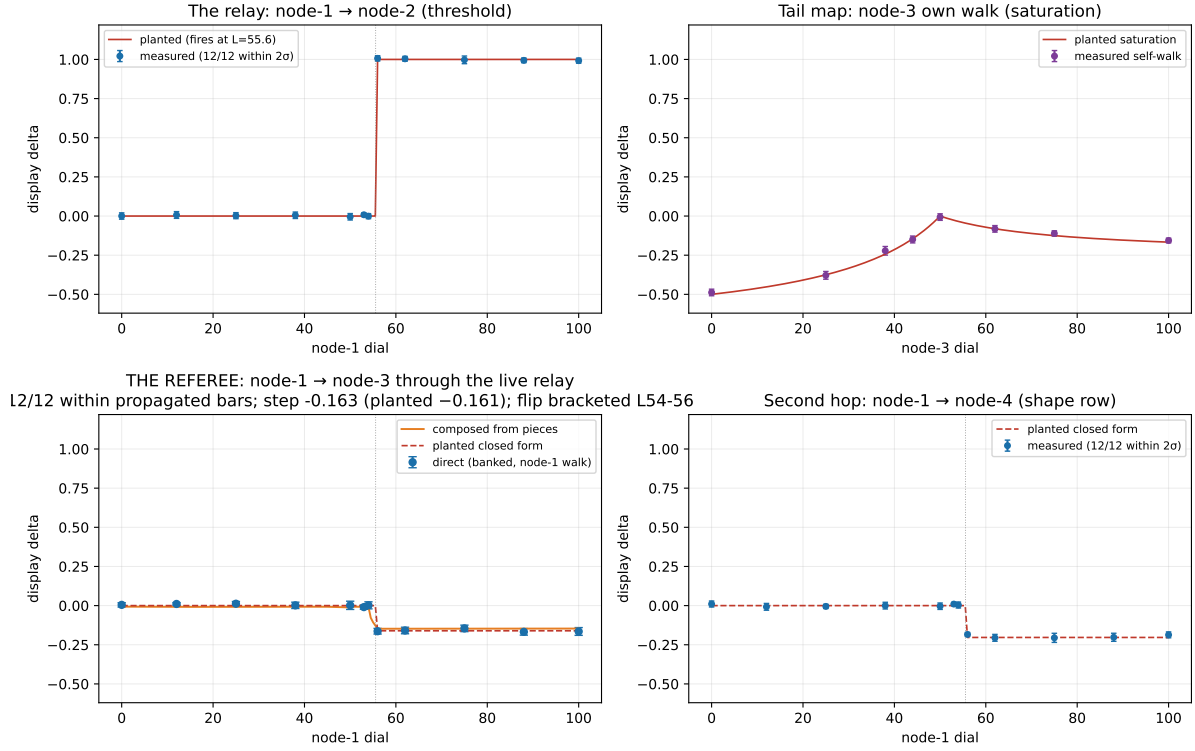


Figure 4: Certification cell, seed 48 (run 33974815211), same design as Figure 3. All panels certify; the flip brackets one level tighter than seed 47.

closed forms, spans thirty and eighteen noise bars. The *salvage* row is a dead plant — authored before the pre-flight gate, its saturation relay can never reach the threshold it was meant to feed ($f_{\text{sat}} \leq 0.5$, the fire line, strictly) — and it returned two clean receipts anyway: the piece matches its closed form at 18/18 points, and with prediction identically zero, all 36 downstream points sit within 2σ of zero. No phantom firing at the discontinuity the author believed in. The *depth boundary* row plants three cascaded nonlinear stages: end-to-end response composes to ≈ 0.001 — twenty times below the floor — and the direct walk measures faithful flat noise. The referee is honestly vacuous there, and the boundary is thereby measured: at $\sigma=0.02$ with this door family, composition reaches ≈ 2 nonlinear hops.

6.6 The metrology of composed bars

The referee rule (2) prices each comparison with propagated bars. Its validation is the campaign's quiet headline. Under the stated rule, 95 of 97 referee points certify (98%; nominal for 2σ is 95.4%), with median deviation $\approx 0.3\sigma$ — the bars are honest and slightly conservative, not tuned. Under the strict variant that ignores the composed side's uncertainty, the count drops to 91/97: the junction row loses three shelf points (z up to 2.97 against predictions whose own bars are $\approx 50\%$ wider than the direct bars there) and cliff seed 48 loses one ($z=2.20$ on a thin direct bar). A referee that punished predictions for uncertainty they had already reported would be over-rejecting; a referee that never missed would be vacuous. The observed behavior sits where honest metrology sits.

7 Boundaries

We name the limits as measured, without softening.

Composition horizon. ≈ 2 nonlinear hops at $\sigma=0.02$ for this door family. Deeper composition needs quieter substrates or linear pipes between doors (which reset the attenuation budget). The horizon is a property of the floor, not of the arithmetic.

Memoryless doors. Every receipt assumes the door answers the same ask the same way — no hysteresis, no state. Doors with memory are the next tier and are untested.

Continuous inputs. Levels are continuous dial positions. Categorical switches (discrete modes) compose differently and are untested.

Static world. The planted substrate is stationary for the cell’s duration. Drift — edges whose strength moves while the map is being built — is the successor question, priced by the map’s refresh economics.

Bench class. All evidence is from a mathematical bench with closed-form truth. No physical substrate appears in this campaign.

Scheduler provenance. Rows before the certification cells ran under earlier protocol builds (timing overrides; pre-ledger scheduling; a retirement verdict that ignored self-curves). Their defects are disclosed with their numbers, and the two certification cells are the rows that carry none of them.

8 Discussion

Why the horizon makes composition necessary. Past two nonlinear hops at this floor, the direct far-end measurement does not become expensive — it becomes *meaningless*: the signal is under the noise and no budget of trials recovers it. The only route to far-end knowledge is through pieces measured where the signal is still strong. Composition is therefore not an optimization of the measurement process; it is the remaining door. The boundary row makes this concrete rather than rhetorical.

The map as computable self-knowledge. A certified, composed coupling map changes what the collecting of agents can do without new measurements: cascades can be traced before anyone fires them (“if this sender pushes, who moves, how much, with what confidence”); attribution questions become lookups; what-if analysis becomes re-simulation on the map, priced by the propagated bars. None of this requires the agents to coordinate beyond the measurement protocol they already run.

Transfer, one paragraph, gated. A map measured on one system is a prior for its neighbors. Whether that prior pays its verification cost is an economics question — map half-life against refresh cost — that drift cells must answer before transfer is claimed. This paper claims nothing there.

Next rung. With detection [8], characterization [7], and composition in hand, the remaining half of the loop is action: using the composed map to steer the system toward chosen operating points, with the same referee discipline (predict, act, film, compare) certifying the steering claims. That is the successor paper’s question, and the composed bars of this one are its entry ticket.

Reproducibility

Every run carries an external identifier (Table 1). The evidence bundle retains the exported curves with per-point bars, the node maps, the coordination trails of the certification cells, and the figure-regeneration script; every figure in this paper regenerates from that bundle by one command. Verdicts are recomputed from curves, never read from run badges.

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